

Hamdard University
Department of Computing

Final Year Project



Smart Medicine Reminder System for DVAGO
(FYP-037/FA25)

Software Design Specifications

Submitted by

Abdul Ghani (2606–2022)
Syed Anas Ali (2640–2022)

Supervisor(s)

Mr. Muhammad Salman
Mr. Maaz Ahmed
Mr. Muhammad Hamza

Spring 2026

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

Document Sign off Sheet

Document Information

Project Title	Smart Medicine Reminder System for DVAGO
Project Code	FYP-037/FA25
Document Name	Software Design Specifications
Document Version	2.0
Document Identifier	FYP-037/FA25-SDS
Document Status	Final Draft
Author(s)	Abdul Ghani, Syed Anas Ali
Approver(s)	Mr. Muhammad Salman, Mr. Maaz Ahmed, Mr. Muhammad Hamza
Issue Date	08/06/2026

Name	Role	Signature	Date
Abdul Ghani	Team Lead		08/06/2026
Syed Anas Ali	Team Member 2		08/06/2026
Mr. Muhammad Salman	Supervisor		08/06/2026
Mr. Maaz Ahmed	Co-Supervisor		08/06/2026
Mr. Muhammad Hamza	External Supervisor		08/06/2026

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

Revision History

Date	Version	Description	Author
18/12/2025	1.0	Initial Draft	Abdul Ghani
08/06/2026	2.0	Final Draft	Abdul Ghani

Definition of Terms, Acronyms, and Abbreviations

Term	Description
FCM	Firebase Cloud Messaging
TTS	Text-to-Speech
OCR	Optical Character Recognition
API	Application Programming Interface
AI	Artificial Intelligence
REST	Representational State Transfer
HTTPS	Hyper Text Transfer Protocol Secure
SSL	Secure Sockets Layer
TLS	Transport Layer Security
MongoDB	Mongo Database
UI	User Interface
JSON	JavaScript Object Notation

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

Table of Contents

Document Sign off Sheet	2
Revision History	3
Definition of Terms, Acronyms, and Abbreviations.....	3
1. Introduction	6
1.1 Purpose of Document	6
1.2 Intended Audience.....	6
1.3 Document Convention	6
1.4 Project Overview.....	7
1.5 Scope	7
2. Design Considerations	8
2.1 Assumptions and Dependencies.....	8
2.2 Risks and Volatile Areas	8
3. System Architecture.....	9
3.1 System Level Architecture	9
3.2 Software Architecture.....	9
4. Design Strategy	10
5. Detailed System Design	11
5.1 Design Class Diagram	11
5.1.1 User	11
5.1.2 Scan	11
5.1.3 Reminder.....	12
5.1.4 NotificationManager.....	12
5.1.5 Key Interactions Between Classes	13
5.2 Database Design.....	14
5.2.1 ER Diagram.....	14
5.2.2 Data Dictionary: User	15
5.2.3 Data Dictionary: Scan	16
5.2.4 Data Dictionary: Reminder	17
5.3 Application Design	18
5.3.1 Sequence Diagram	18
5.3.2 State Diagram	20

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5.3.3 GUI Design..... 21

6.0 References 23

Appendices..... 23

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

1. Introduction

This document lays out the Software Design Specification (SDS) for the Smart Medicine Reminder System being built for DVAGO. The system helps people stay on top of their medication by combining scheduled reminders, push and voice alerts, and AI-assisted medicine recognition through image scanning. What follows captures the architectural choices, the structure of the system, and the detailed design decisions that the development team is using to build the solution.

1.1 Purpose of Document

This document exists to give a clear, detailed account of how the Smart Medicine Reminder System for DVAGO is designed, its architecture, its components, and the reasoning behind major design decisions. It is meant to be a reference point that developers, supervisors, and evaluators can return to throughout the project to understand how the system is structured and why it was built that way.

1.2 Intended Audience

This document is written for the project supervisor and co-supervisor, the Final Year Project evaluation committee, the members of the development team, and, where relevant, DVAGO's technical staff. It is intended to give each of these groups enough detail to follow the system's design, track its progress against project objectives, and support review and approval at each stage of the project.

1.3 Document Convention

To keep the document consistent and easy to follow, the following conventions are used throughout:

- **Font:** Calibri
- **Font Size:** 12 pt for body text, 14 pt for subsection titles, 16 pt for main section titles

Formatting rules:

- Bold text is used for section headings
- Italics are used for emphasis
- Numbered lists are used for steps that happen in sequence
- Bullet points are used for descriptive, non-sequential information

1.4 Project Overview

The Smart Medicine Reminder System for DVAGO is a mobile application that lets users set up medicine reminders and get timely alerts so they don't miss a dose. Its core features include:

- Secure registration and login for users

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

- Medicine reminders that can be scheduled by time and frequency
- Push notifications and voice alerts at the scheduled time
- AI-assisted recognition of medicine names through image scanning
- A history of past reminders that users can look back on

The system is built around a layered architecture: a mobile front end, a set of backend services, and cloud-based storage. Firebase handles authentication, notification delivery, and a good part of the backend integration work, which keeps the system scalable without adding unnecessary complexity.

1.5 Scope

In Scope

- Building a mobile application that runs on both Android and iOS
- User authentication through Firebase
- Scheduling and managing medicine reminders
- Delivering push notifications and voice alerts
- Recognizing medicine names from scanned images using AI
- Storing and retrieving a history of reminders

Out of Scope

- Online ordering of medicine or any payment processing
- Medical diagnosis or validation of prescriptions
- Direct integration with DVAGO's live production systems
- Live chat or consultation features with pharmacists

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

2. Design Considerations

This section walks through the assumptions, dependencies, and risks that shaped the design of the system. Thinking through these ahead of time helps keep the design realistic and gives the team a way to adapt if circumstances change during development.

2.1 Assumptions and Dependencies

Assumptions

- Users own an Android or iOS smartphone capable of running the app
- Users are willing to grant the permissions the app needs: notifications, camera access, and audio output
- A working internet connection is available whenever the app needs to reach cloud services
- Users are mainly engaging with the app to manage their medicine reminders, rather than for unrelated purposes

Dependencies

- Firebase services need to stay available and reliable, since authentication and notifications both run through them
- The Space OCR API needs to perform reasonably well at reading text off medicine packaging
- Push notification delivery depends on the underlying mobile operating system cooperating with Firebase Cloud Messaging

2.2 Risks and Volatile Areas

- **Notification Failure:** Some devices delay or block notifications depending on their battery-saving or permission settings. Mitigation: rely on Firebase Cloud Messaging's retry behaviour to reduce the chance of a missed alert.
- **AI Scan Accuracy:** Poor lighting or a blurry photo can throw off the OCR result. Mitigation: keep manual entry available as a fallback whenever a scan doesn't come back clean.
- **Technology Changes:** Third-party APIs evolve, and a breaking change upstream could affect the app. Mitigation: keep the design modular and pin API versions where possible, so updates don't ripple through the whole system.
- **User Adoption:** If the app feels complicated, less tech-savvy users may give up on it. Mitigation: keep the interface simple and avoid burying core actions behind extra steps.

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

3. System Architecture

This section steps back and looks at how the major pieces of the system are organized and how they work together to deliver the app's functionality.

3.1 System Level Architecture

The system breaks down into four main components:

- **Mobile Application Layer:** the interface users interact with to manage reminders and run scans
- **Backend Services Layer:** handles authentication, reminder scheduling, and the logic behind notifications
- **Cloud Services Layer:** Firebase Authentication, Firebase Cloud Messaging, and MongoDB
- **OCR Processing Layer:** Space OCR API, used to read medicine names off wrapper images

3.2 Software Architecture

The software itself follows a layered structure:

- **User Interface Layer:** the mobile UI that users see and tap through
- **Application Logic Layer:** schedules reminders, validates input, and triggers notifications
- **Data Access Layer:** stores and retrieves user data and reminder history
- **Integration Layer:** talks to Firebase and other external APIs

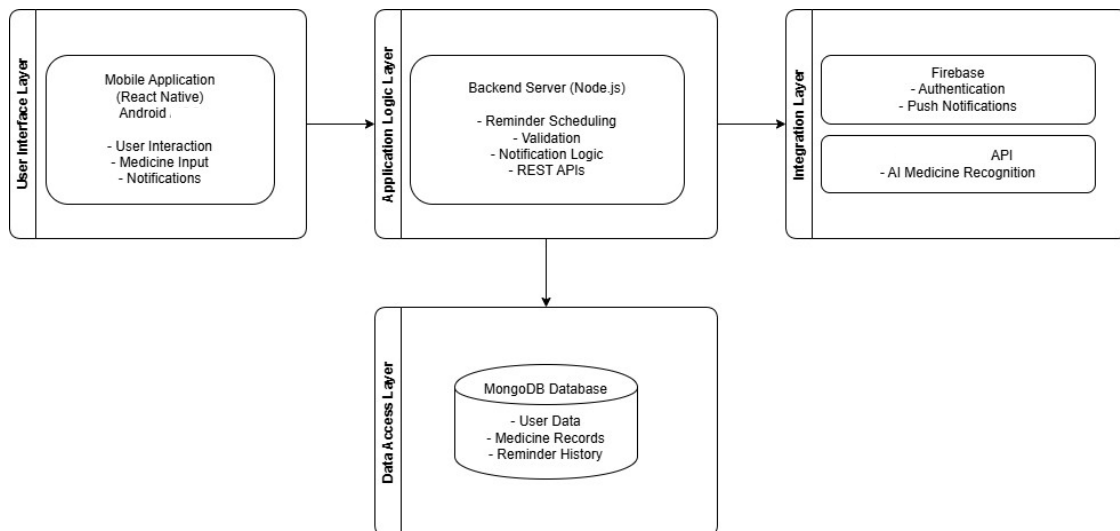


Figure 1: Software Architecture for the Proposed System

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

4. Design Strategy

The design strategy behind the Smart Medicine Reminder System for DVAGO is built around four priorities: keeping things simple, modular, reliable, and easy to extend later. The team leaned on Object-Oriented Design principles throughout and chose cloud-based services specifically because they make the architecture easier to maintain and scale as the app grows.

Future System Extension or Enhancement

- **Strategy:** Keep the architecture modular, with reminders, notifications, and AI scanning built as separate, independent components.
- **Reasoning:** This makes it possible to bolt on features later to admin dashboards, usage analytics, or wearable device integration without having to touch the core of the app.
- **Trade-off:** It takes more upfront planning to design things this way, but it pays off when the app needs to grow.

System Reuse

- **Strategy:** Build authentication, notification handling, and data storage as reusable modules rather than one-off implementations.
- **Reasoning:** Less duplicated code means fewer places for bugs to hide and an easier time maintaining the app long-term.
- **Trade-off:** Generalizing these components takes a bit more effort up front than writing them for a single use case.

User Interface Paradigms

- **Strategy:** Design a mobile-first, minimalistic interface.
- **Reasoning:** A simpler interface asks less of the user and works better across different age groups and comfort levels with technology.
- **Trade-off:** Fitting everything onto a smaller, cleaner screen means being deliberate about what gets shown and what gets tucked away.

Data Management (Storage, Distribution, Persistence)

- **Strategy:** Use Firebase's cloud-based NoSQL storage for persisting data.
- **Reasoning:** It keeps data safe, syncs in real time, and makes sure reminders carry over correctly between sessions and devices.
- **Trade-off:** Some queries that would be straightforward in a relational database take more care to get right in a NoSQL setup.

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

Concurrency and Synchronization

- **Strategy:** Handle notifications and other background work asynchronously.
- **Reasoning:** This keeps the app responsive, and reminders get delivered on schedule without freezing up the interface while that's happening.
- **Trade-off:** Asynchronous code is inherently a bit more complex to implement and test than a purely sequential approach.

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5. Detailed System Design

5.1 Design Class Diagram

This section walks through the first pass at the class diagram for the system, covering the core entities and how they relate to one another. It gives a structural view of the system as it stands today, and it's expected to be refined further as implementation moves forward and requirements get clearer.

5.1.1 User

Description:

Represents the end user, the patient, who uses the app to manage their medicine reminders and notification preferences.

Attributes:

- user_id: a unique identifier for the user
- name: the user's full name
- email: the email address used to register
- password: an encrypted password used for authentication
- fcmToken: the Firebase Cloud Messaging token tied to the user's device

Methods:

- register(): creates a new user account
- login(): authenticates the user against backend services
- updateProfile(): updates the user's stored profile information

Interactions:

- A User can create multiple Reminders
- A User can perform multiple medicine Scans

5.1.2 Scan

Description:

Represents an AI-assisted scan of a medicine wrapper, used to pull medicine information out of an image.

Attributes:

- scan_id: a unique identifier for the scan
- medicineName: the medicine name detected from the scan
- extractedText: the raw text pulled out by OCR

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

- imageUrl: where the scanned image is stored
- createdAt: when the scan was taken

Methods:

- scanImage(): processes the captured image to extract text
- saveScan(): stores the result of the scan

Interactions:

- Each Scan belongs to one User

5.1.3 Reminder

Description:

Handles the scheduling and tracking of medicine intake reminders, including whether notifications and voice alerts are turned on for that reminder.

Attributes:

- reminder_id: a unique identifier for the reminder
- medicineName: the name of the medicine being tracked
- dosage: dosage instructions for that medicine
- time: the time the reminder is scheduled to fire
- frequency: how many times per day the reminder repeats
- duration: how many days the reminder should stay active
- notification: whether push notifications are enabled for this reminder
- voiceAlert: whether voice alerts are enabled for this reminder
- notified: whether the notification has already been sent

Methods:

- scheduleReminder(): creates a new reminder schedule
- updateReminder(): changes the timing or details of an existing reminder
- cancelReminder(): deactivates a reminder

Interactions:

- Each Reminder belongs to one User
- A Reminder triggers notifications through the NotificationManager

5.1.4 NotificationManager

Description:

A service class responsible for managing both push and voice notifications based on each reminder's schedule.

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

Attributes:

- notification_id: a unique identifier for the notification
- type: whether the notification is push or voice
- delivery_status: whether the notification was sent successfully or failed

Methods:

- sendPushNotification(): sends an alert through Firebase Cloud Messaging
- playVoiceAlert(): triggers a voice alert on the device
- scheduleNotification(): sets up when a notification should fire

Interactions:

- Receives trigger events from a Reminder
- Delivers the resulting alert to the User

5.1.5 Key Interactions Between Classes

- **User and Reminder** — a User can create, update, or delete any number of Reminders.
- **User and Scan** — a User can run multiple medicine scans to pull out medicine information.
- **Reminder and NotificationManager** — Reminders are what trigger push or voice notifications at the scheduled time.
- **User and NotificationManager** — notifications ultimately get delivered according to the user's own preferences and device settings.

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

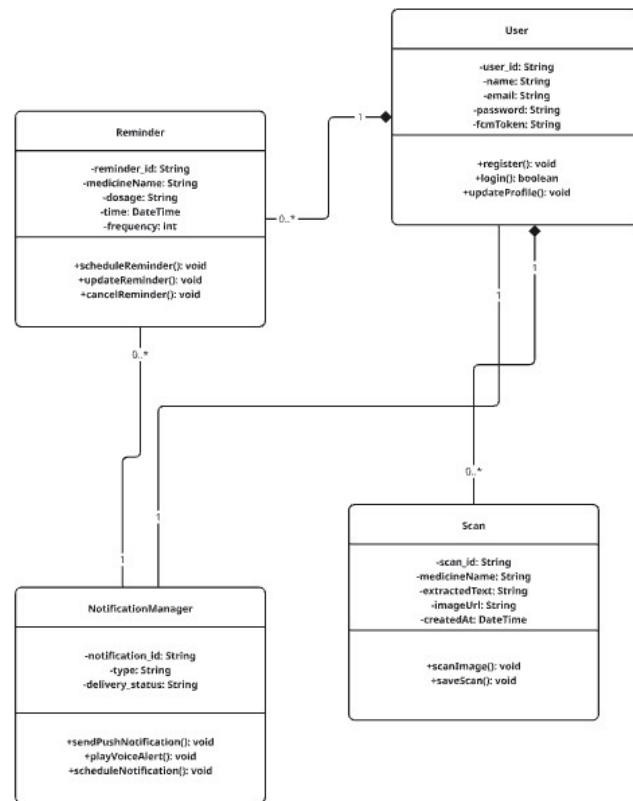


Figure 2: Initial Class Diagram for the Smart Medicine Reminder System

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5.2 Database Design

The database follows a logical Entity Relationship (ER) model, implemented on a cloud-based MongoDB instance. This gives the system flexible document storage, room to scale, and real-time synchronization — all useful for a mobile app that needs reminders and scan data to stay in sync across sessions.

Entities

- User
- Reminder
- Scan

Relationships

- A User can create multiple Reminders
- A User can perform multiple Scans
- Each Reminder belongs to one User
- Each Scan belongs to one User

5.2.1 ER Diagram

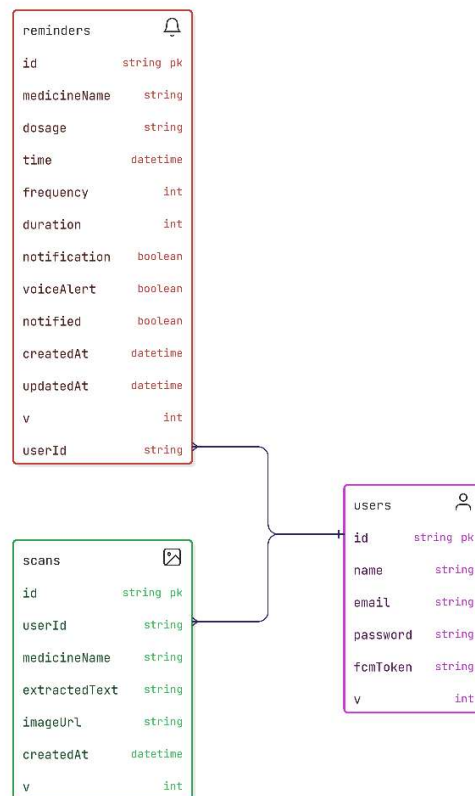


Figure 3: ER Diagram for the Smart Medicine Reminder System

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5.2.2 Data Dictionary: User

5.2.2.1 Data 1: User

Field	Detail
Name	User
Alias	Patient, App User
Where-used / How-used	Used for authentication, identifying the user, and linking reminders and scan records (input, store).
Content Description	user_id = name + email + password + fcmToken

5.2.2.2 Column Details

Column	Description	Type	Length	Nullable	Default Value	Key
_id	Unique user identifier	ObjectId	12 bytes	No	Auto-generated	PK
name	User full name	String	255	No	NULL	
email	User email address	String	255	No	NULL	
password	Encrypted password (hashed)	String	255	No	NULL	
fcmToken	Firebase Cloud Messaging token	String	255	Yes	Auto-generated	
__v	Document version	Number	N/A	Yes	0	

5.2.3 Data Dictionary: Scan

5.2.3.1 Data 2: Scan

Field	Detail
Name	Scan
Alias	Medicine Scan / Wrapper Scan
Where-used / How-used	Used to store AI-extracted medicine text and the associated scanned image data (input, store).
Content Description	scan_id = userId + medicineName + extractedText + imageUrl + {createdAt}

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5.2.3.2 Column Details

Column	Description	Type	Length	Nullable	Default Value	Key
_id	Unique scan identifier	ObjectId	12 bytes	No	Auto-generated	PK
userId	Reference to the User who ran the scan	ObjectId	12 bytes	No	NULL	FK
medicineName	Detected medicine name	String	255	Yes	NULL	
extractedText	OCR-extracted text	String	N/A	Yes	NULL	
imageUrl	Stored image path	String	500	No	NULL	
createdAt	Scan timestamp	Timestamp	N/A	No	CURRENT_TIMESTAMP	
__v	Document version	Number	N/A	Yes	0	

5.2.4 Data Dictionary: Reminder

5.2.4.1 Data 3: Reminder

Field	Detail
Name	Reminder
Alias	Medicine Reminder, Alert Schedule
Where-used / How-used	Used to manage medicine schedules and to trigger push or voice notifications (store, process).
Content Description	reminder_id = userId + medicineName + dosage + time + frequency + duration + notification + voiceAlert + notified + {createdAt}

5.2.4.2 Column Details

Column	Description	Type	Length	Nullable	Default Value	Key
_id	Unique reminder identifier	ObjectId	12 bytes	No	Auto-generated	PK
userId	Reference to the User who owns the reminder	ObjectId	12 bytes	No	NULL	FK
medicineName	Medicine name	String	255	No	NULL	

Smart Medicine Reminder System for DVAGO					Version: 2.0	
Software Design Specifications					Date: 08/06/2026	
FYP-037/FA25-SDS						
Column	Description	Type	Length	Nullable	Default Value	Key
dosage	Dosage instructions	String	100	No	NULL	
time	Reminder date and time	DateTime	N/A	No	NULL	
frequency	Times per day	Number	N/A	No	1	
duration	Duration in days	Number	N/A	No	NULL	
notification	Whether push notification is enabled	Boolean	N/A	Yes	true	
voiceAlert	Whether voice alert is enabled	Boolean	N/A	Yes	false	
notified	Whether the notification has been sent	Boolean	N/A	Yes	false	
createdAt	Creation timestamp	Timestamp	N/A	No	CURRENT_TIMESTAMP	
updatedAt	Last update timestamp	Timestamp	N/A	No	CURRENT_TIMESTAMP	
__v	Document version	Number	N/A	Yes	0	

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5.3 Application Design

This section gets into the flow of interactions and state transitions that drive the app. The sequence and state diagrams below, along with the explanations that go with them, cover the key processes involved in setting up a medicine reminder.

5.3.1 Sequence Diagram

Actors:

- End User

Modules / Components:

- **Mobile Application (React Native UI):** where users enter or scan medicine details.
- **Camera Module:** captures images of medicine strips or prescriptions.
- **OCR Module (Space OCR API):** extracts text from the captured image using AI-based optical character recognition.
- **Backend Server (Node.js API):** handles business logic, validation, and request processing.
- **MongoDB Database:** stores reminder data persistently.
- **Firebase Cloud Messaging (FCM):** schedules and delivers reminder notifications.

Main Processes (left to right):

1. End User — launches the app, picks either Manual Entry or AI Scan, enters the medicine name, dosage time, and frequency (or scans the medicine with the camera), confirms the extracted details if it was a scan, and gets a confirmation once the reminder is created.
2. Mobile Application — shows the right input form or activates the camera, sends the entered or extracted data to the backend, and shows a confirmation or error message depending on what comes back.
3. Backend Server — receives the reminder data, validates the medicine information and scheduling details, stores the reminder, and triggers notification scheduling through Firebase Cloud Messaging.
4. Database (MongoDB Atlas) — saves the validated reminder record and keeps it available for retrieval later when it's time to fire the notification.
5. Firebase Cloud Messaging (FCM) — schedules the push or voice notification based on the dosage time and delivers the alert when that time comes.

Flow of Communication:

- Every action the user takes sets off a chain of operations across the different components: the mobile app collects the input or scanned data, the backend validates and processes the request, the database stores it securely, and Firebase Cloud Messaging takes care of scheduling and delivering the notification.
- The system is designed so that AI-based text extraction, backend processing, and notification delivery all stay in sync with one another.

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

General Purpose:

These two diagrams show how the end user interacts with the reminder system and how the different components work together to create and manage reminders — covering both the manual entry path and the AI-scan path, from data input through validation, storage, and notification scheduling. Together they highlight how AI, backend processing, and cloud notifications combine to keep reminders accurate and on time.

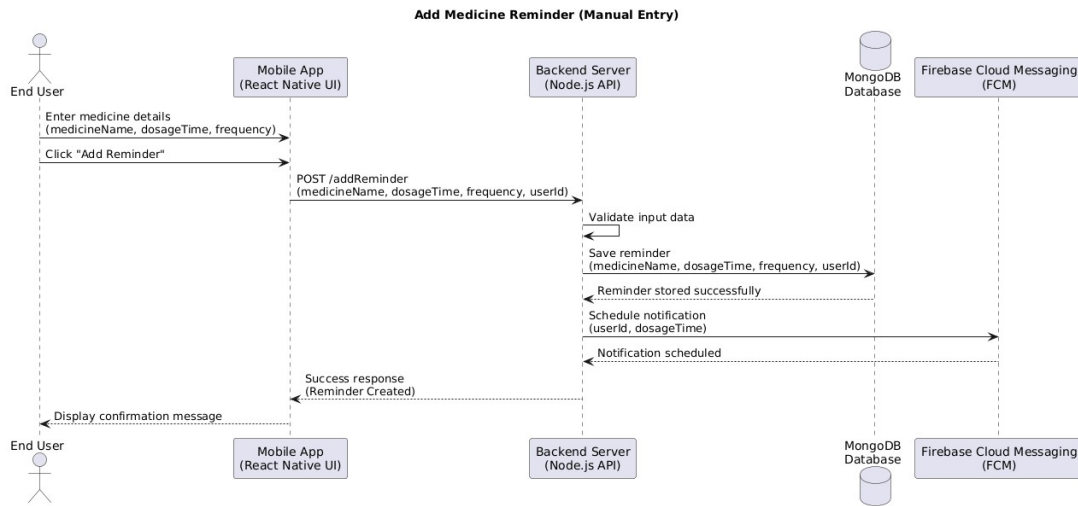


Figure 4: Sequence Diagram — Add Medicine Reminder (Manual Entry)

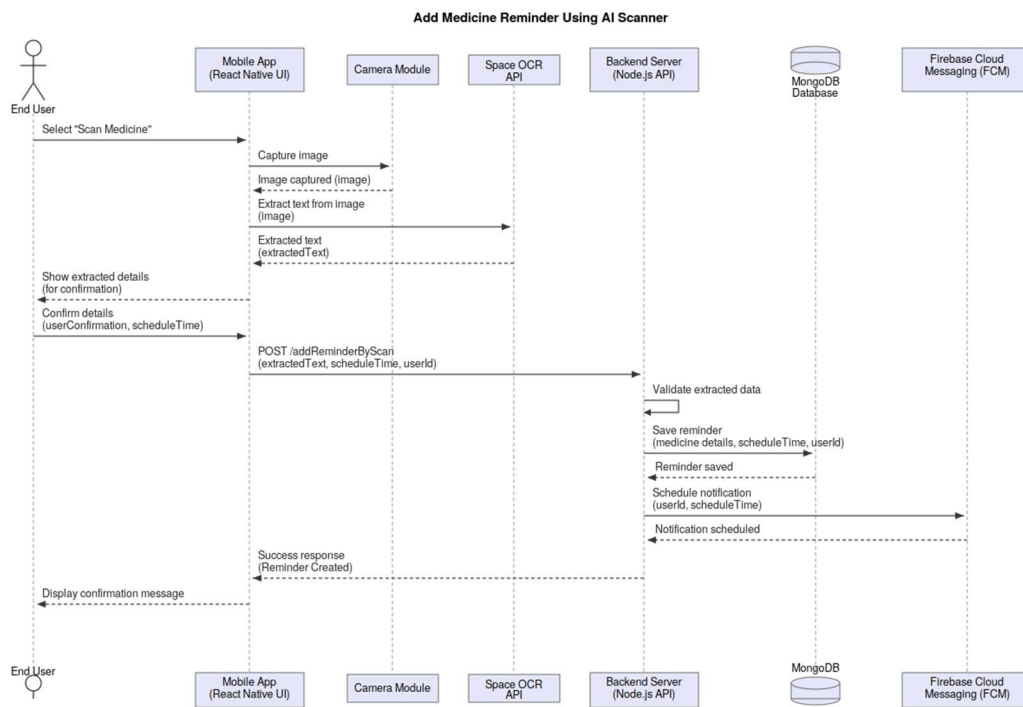


Figure 5: Sequence Diagram — Add Medicine Reminder Using the AI Scanner

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5.3.2 State Diagram

Components:

1. Medicine Reminder System (Central System)
 - Reminder Management Module — handles creating, updating, and deleting reminders, whether they were entered manually or pulled from an AI scan.
 - User Management Module — manages user profiles, authentication, and each user's own reminder data.
 - AI Scanning & Processing Module — captures medicine images, runs the Space OCR API to extract text, and lets the user confirm what was detected before it gets saved.
 - Notification Scheduling Module — coordinates reminder schedules and fires alerts through Firebase Cloud Messaging.
 - Backend Processing Module — validates data, applies business logic, and manages the communication between the app, the database, and the notification service.
2. Data Storage and Retrieval
 - Backs the rest of the system by storing and retrieving user profiles, medicine details, dosage schedules, and the reminder/notification history.

Interaction Flow:

- **End User:** adds reminders manually through the Reminder Management module, or scans medicine using the AI Scanning & Processing module, confirms the extracted details, and then receives notifications through the Notification Scheduling module.
- **Mobile Application:** sits between the user and the system, sending reminder data to the backend for validation and storage, and showing confirmation messages and alerts back to the user.
- **Backend System:** processes incoming reminder requests, coordinates with the database for storage and retrieval, and talks to Firebase Cloud Messaging to schedule and deliver notifications.

Data Flow:

- Data moves between the Medicine Reminder System and the Data Storage and Retrieval component to store and pull medicine reminder details, scheduling information, and user-specific preferences.
- The system keeps AI processing, backend validation, database storage, and notification delivery synchronized with one another.

Purpose:

This diagram lays out the functional architecture of the app and shows how its different parts work together to manage reminders. It highlights the modular design, the centralised handling of data, and how AI-based scanning and cloud notifications fit into the bigger picture — all in service of getting reminders created, scheduled, and delivered accurately.

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

Initial State Diagram for Proposed Smart Medicine Reminder System

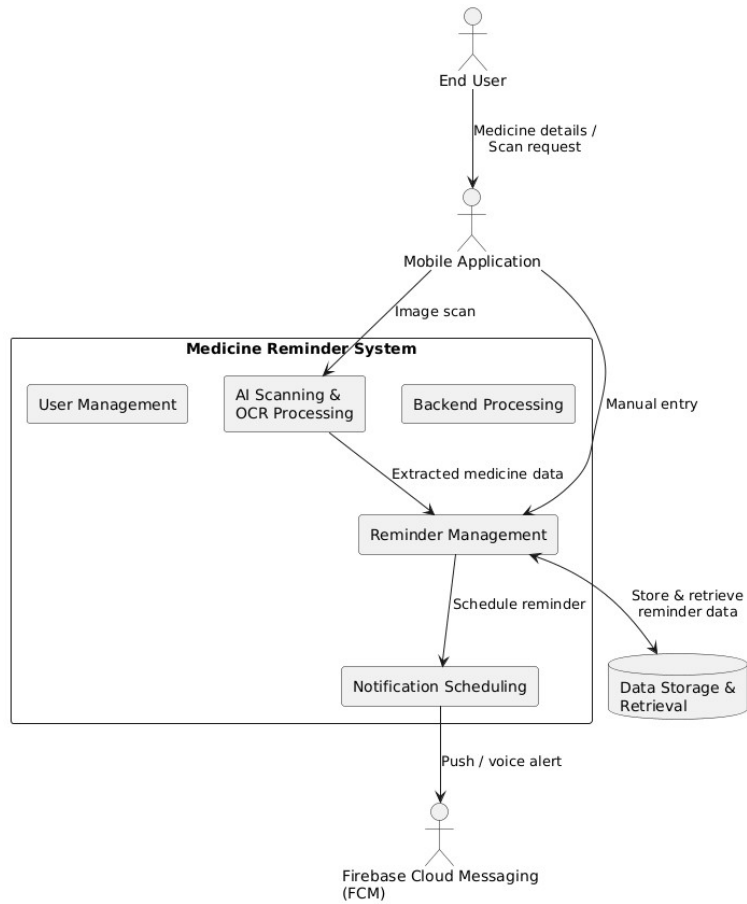


Figure 6: Initial State Diagram for the Proposed Smart Medicine Reminder System

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

5.3.3 GUI Design

The screens shown below represent an early pass at the GUI, a working starting point rather than a finished design. They reflect the team's current understanding of what users need and how the app should function, and they're expected to change as feedback comes in and the design is iterated on. Getting this right matters: the GUI needs to stay user-friendly, efficient, and genuinely useful for the workflows it's meant to support.

- Splash → Tutorial → Login → Signup
- Dashboard → My Medicine → Analytics
- Add Medicine → Success Prompt → Scan Medicine
- Profile → Log Out

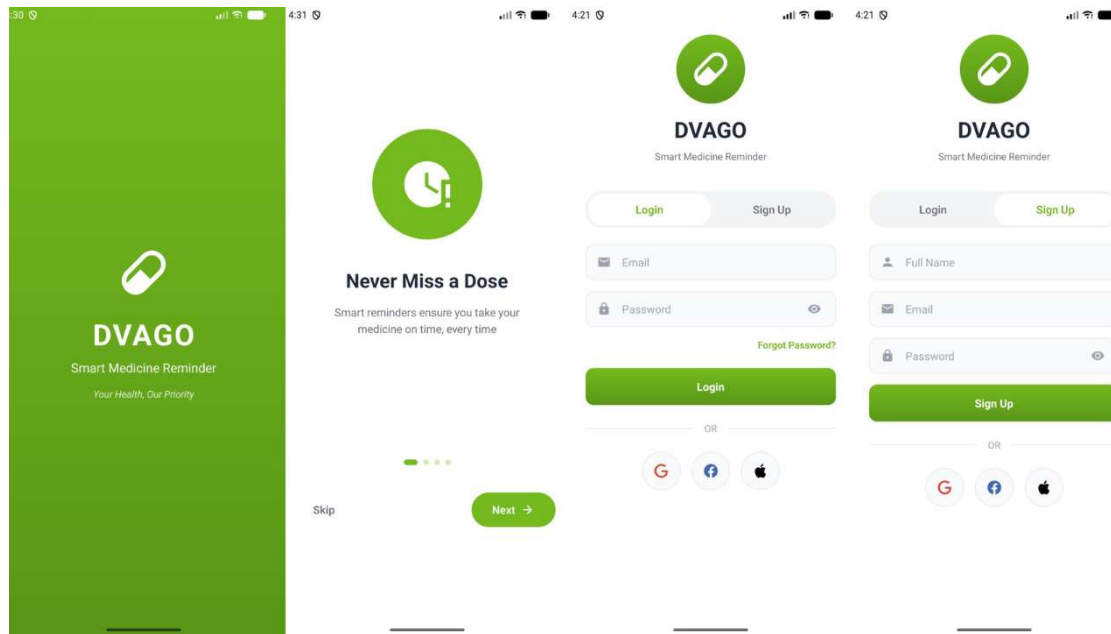


Figure 7: Splash, Onboarding, Login, and Sign-Up Screens

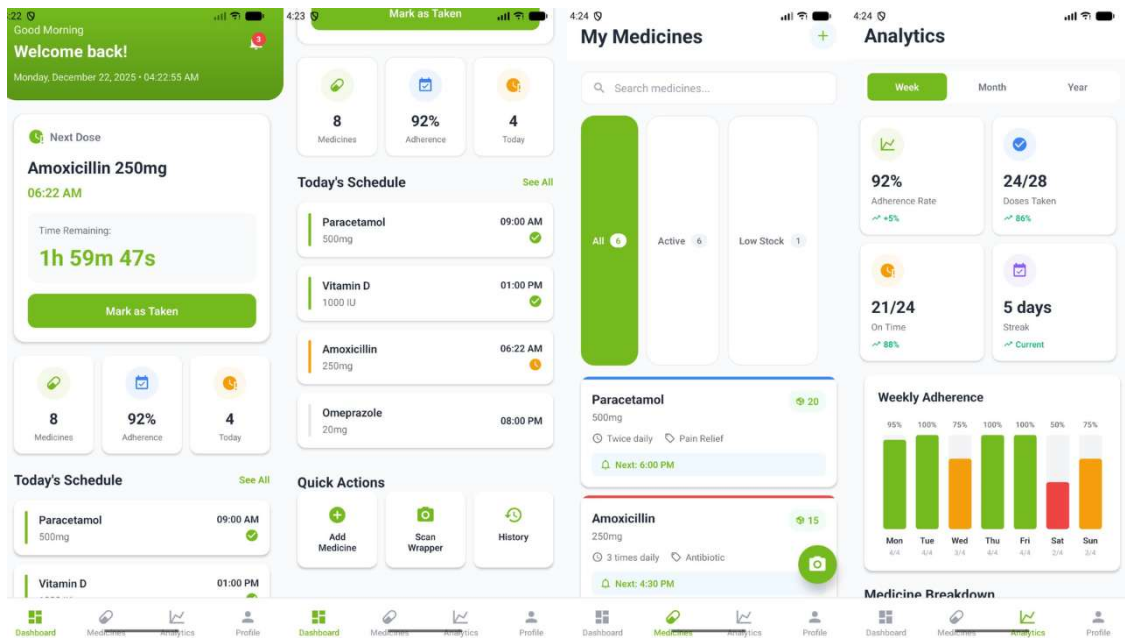


Figure 8: Dashboard, My Medicines, and Analytics Screens

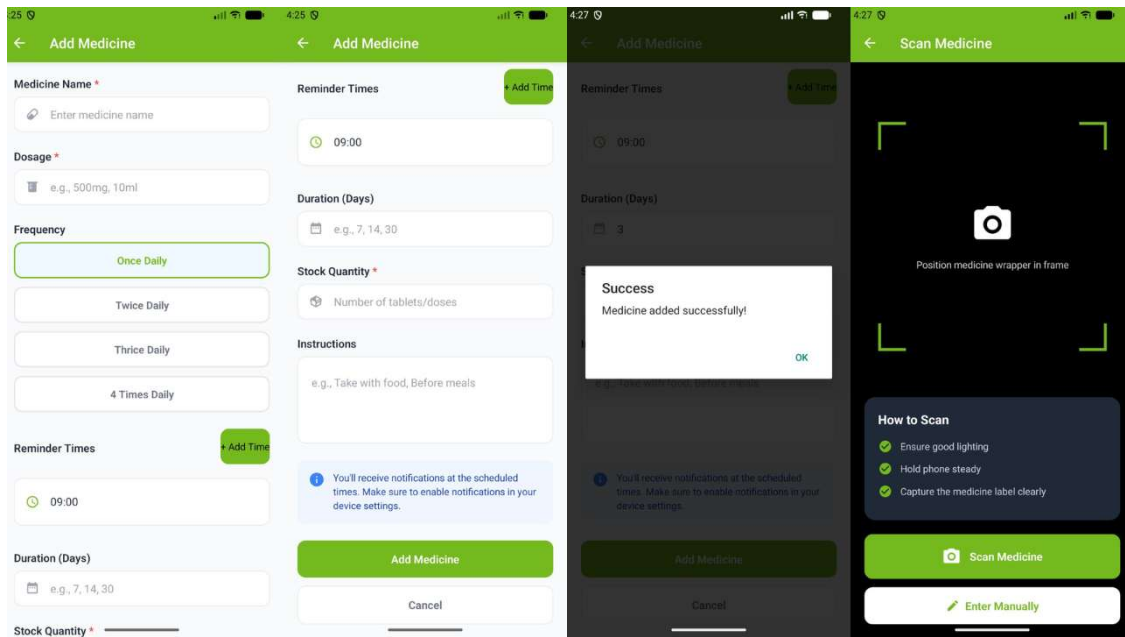


Figure 9: Add Medicine and Scan Medicine Screens

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

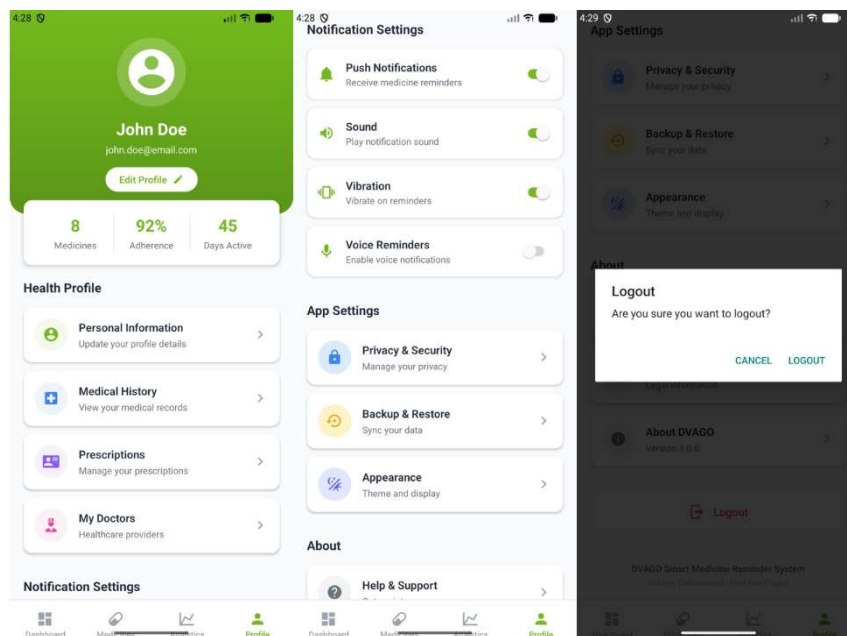


Figure 10: Profile, Notification Settings, and App Settings Screens

Smart Medicine Reminder System for DVAGO	Version: 2.0
Software Design Specifications	Date: 08/06/2026
FYP-037/FA25-SDS	

6.0 References

1. "Improving Medication Adherence with Mobile Apps," JMIR, 2022.
2. Firebase Documentation: Cloud Messaging.
3. Web Speech API Documentation.
4. Medisafe App — Features Overview.
5. World Health Organization, Report on Digital Health in Developing Countries.
6. Morrison, A. et al., Journal of Medical Internet Research, 2021.
7. World Health Organization, Adherence to Long-term Therapies, 2019.
8. Medisafe — <https://www.medisafe.com>
9. MyTherapy — <https://www.mytherapyapp.com>

Appendices

No additional appendices are included in this version of the document. Supporting material wireframes, extended data samples, and supervisor feedback notes will be appended here as the design is refined in later revisions.